

StitchAI

A Zero-Shot/Few-Shot Vision-Language Framework for Unified Defect, Safety, and Machinery Anomaly Detection in Bangladesh's Garment Sector

Team: HexaMind | United International University

Members: Muhaiminul Rashid Shad - Team Lead, Abrar Jahin, Sumaya Zaman, Md. Shahriar Morshed, Md Ahnaf Tahmid, Nusrat Jahan Moon

1 Vision and Problem Statement

1.1 The Problem

Bangladesh's Ready-Made Garments (RMG) sector is a major part of the national economy, accounting for 80.62% of the country's total exports in FY 2025–26 [1]. Because of this scale, consistent product quality, workplace safety, and reliable factory operations are important not only for individual manufacturers but also for buyer confidence and the competitiveness of the sector.

Within a garment factory, these areas involve different inspection needs. Quality-control teams look for defects in fabrics and garments, safety personnel monitor workplace conditions, and maintenance teams inspect equipment for visible signs of wear or damage. AI can support these activities, but the conventional approach is usually to develop a separate task-specific model for each problem. This means collecting and labelling new data, retraining models, and maintaining multiple systems whenever a new defect, product type, or inspection task is introduced.

The challenge is therefore broader than detecting one specific defect. The key question is whether a shared visual AI framework can support several selected RMG inspection tasks while reducing the amount of task-specific data and retraining required.

1.2 Vision and Proposed Solution

We propose a Universal Anomaly Detection Framework for Bangladesh's RMG sector. The framework will use a shared vision-language and anomaly-detection backbone for tasks where this approach is suitable. For anomalies that require more specific object or scene understanding, such as certain safety conditions, a lightweight task-specific visual component can be added without changing the overall inspection workflow.

The prototype will focus on three areas:

- **Garment and fabric quality inspection:** Identifying visible defects such as stains, holes, stitching problems, misalignment, or other surface abnormalities.
- **Visual safety screening:** Identifying selected safety-related conditions that can be reliably recognised from images.
- **Visible machinery anomaly screening:** Identifying unusual wear, surface damage, or abnormal appearance of machine components. The system will not attempt to diagnose internal mechanical or

electrical faults from an external image.

Recent work such as WinCLIP and AnomalyCLIP has shown that vision-language models can perform zero-shot or few-shot anomaly detection across diverse visual categories [2][3]. These methods motivate our low-data approach, although the project will evaluate their suitability separately for each selected RMG task rather than assuming that one method performs equally well in every domain.

For each input image, the system will return a confidence or anomaly score and, where possible, highlight the suspicious region. A short explanation will help the supervisor understand the result. The AI will act as a decision-support tool. But the final quality, safety, or maintenance decisions will still remain with trained personnel.

The central research question is: Can a shared visual framework achieve useful performance across selected RMG scenarios with lower data and retraining requirements than separate task-specific models?

The prototype will compare the shared approach with relevant conventional baselines using measures such as detection performance, false-negative rate, localisation quality, inference time, and the amount of task-specific data required. This is a deliberate adaptation of a real, published line of research (CLIP-based zero-shot/few-shot industrial anomaly detection, see Section 2.2) to a domain – Bangladeshi RMG manufacturing – where it has not previously been applied, rather than an attempt to invent new model architecture from scratch. The full technical design is set out in Section 4.

1.3 Relevance to Contemporary Needs

The project is relevant to several current needs of Bangladesh’s RMG sector. First, the industry’s large contribution to national exports makes efficient and consistent quality inspection economically important [1]. Second, workplace safety remains an active operational concern. The RMG Sustainability Council continues to conduct factory safety programmes, remediation monitoring, inspections, training, and occupational-safety complaint handling in Bangladesh [4]. Third, factories may need to work with different products, materials, production environments, and inspection requirements. Developing a completely separate AI system for every new visual task can increase data, training, and maintenance requirements. Recent zero-shot and few-shot research offers an opportunity to investigate whether some of this burden can be reduced [2][3]. The project therefore does not claim that a single AI model can solve every factory problem. Its aim is more specific: to test whether shared visual intelligence and low-data adaptation can make AI-based inspection more flexible across selected RMG tasks.

2 Use Case and Existing Solutions

2.1 Primary Use Cases

The framework is designed around four concrete factory-floor use cases, each with an identified end user and an identified decision it supports. In every case the system produces a flag, a score, and a localisation for a human to act on; it does not act autonomously.

- **Inline quality control** (QC staff, line supervisors). A camera at a fabric inspection station or roll-to-roll line feeds images to the system, which flags stains, holes, stitching faults, and other visible surface abnormalities for a human inspector to confirm before the roll or garment moves downstream. The benefit is earlier detection in the production sequence, where rework is cheapest.
- **Visual safety screening** (safety officers, compliance staff). Images or sampled CCTV frames are screened for selected safety-related conditions that are reliably recognisable from an image, with flagged frames routed to the safety officer for review. This supports both day-to-day safety management and the evidence packs factories must assemble for buyer safety audits.
- **Visible machinery anomaly screening** (maintenance teams) – proof of concept. Images of machine components are compared against ‘normal’ reference states to flag visible wear, surface damage, or

abnormal appearance before it develops into a stoppage. As set out in Section 3.2, this category is demonstrated in the prototype using proxy industrial-anomaly data, with real RMG-machinery data collection defined as the next step.

- **Compliance and audit reporting** (factory management, buyer auditors). Every inference is logged with its category, score, verdict, and timestamp, producing a single consolidated, timestamped record that spans all three anomaly types. Today these three inspection activities generate three separate paper or spreadsheet trails, if they generate one at all; a unified log is a direct output of routing everything through one pipeline.

The common thread is that all four use cases are served by the same deployed pipeline and the same interface. Adding a fourth or fifth inspection task later means supplying a new bank of reference images, not commissioning a new model, new labelling effort, and new maintenance overhead.

2.2 Existing Solutions and Gap Analysis

The table below compares the categories of solutions currently available against our proposed approach.

Existing Solution Category	What It Offers	Limitation vs. Our Approach
Industrial fabric inspection hardware (e.g., Uster, Barco-class systems)	High-accuracy, purpose-built fabric defect scanning	Expensive, foreign, single-purpose – cannot extend to safety or machinery without separate systems
Western construction-site PPE detection products	CV-based PPE/safety compliance monitoring	Not localized to RMG factories, single-purpose, not designed for fabric or machinery anomalies
Published zero-/few-shot anomaly research (WinCLIP, AnomalyCLIP, AnomalyGPT)	Category-agnostic anomaly detection methodology, validated on general benchmarks	Research-stage code, not adapted or deployed for any specific industry – no RMG application, no explanation-layer product, no deployment pipeline exists today
Typical student-project CV classifiers	Single-category defect or safety classifiers (e.g., one CNN per task)	Cannot generalize across anomaly types without training a new model per category – the exact fragmentation problem we are solving

Table 1: Existing solution categories and their limitations relative to the proposed framework.

The gap our project fills is the intersection of these four: a single deployable pipeline that (1) generalizes across anomaly categories via zero-/few-shot VLM methodology, (2) produces human-readable explanations rather than a bare score, (3) is explicitly adapted to the RMG domain rather than left as generic research code, and (4) is deployable on free-tier infrastructure accessible to resource-constrained factories rather than only large enterprises.

Each of the four rows above satisfies at most two of these conditions. Purpose-built inspection hardware is accurate and deployable but neither general nor affordable at the scale of a mid-sized Bangladeshi factory. Published research is general and current but is not a product: it ships as benchmark code with no domain adaptation, no explanation layer, and no serving path. Single-task classifiers are cheap to build but reproduce the fragmentation this project exists to remove. No existing option occupies all four positions at once, and that intersection is the contribution being claimed – an adaptation and integration contribution, not a claim to have invented a new model architecture.

2.3 Scope, Objectives, and Expected Outcomes

In scope. A single shared inference pipeline covering the three categories named in Section 1.2; a few-shot reference bank per category; a natural-language explanation layer over detected anomalies; a unified inference log supporting audit reporting; and a working demonstration interface with a public code repository.

Out of scope (deliberately). Internal mechanical or electrical fault diagnosis from external imagery; autonomous rejection, shutdown, or disciplinary action; real-time multi-camera video analytics at full production frame rates; and any claim of validated accuracy on real Bangladeshi factory imagery, which requires the pilot partnership described in Section 3.1.

Objectives. (i) Demonstrate anomaly detection across at least two, ideally three, RMG-relevant categories through one shared backbone; (ii) attach a plain-language explanation to each flagged anomaly; (iii) quantify the trade-off between the shared approach and conventional task-specific baselines on detection performance, false-negative rate, localisation quality, inference time, and the volume of task-specific data required; and (iv) keep the entire prototype runnable on free-tier infrastructure.

Expected outcomes. A deployable inference model and public repository meeting the competition's prototype requirement; measured per-category results reported honestly alongside their disclosed limitations; and a documented, low-risk path from the prototype to a single-factory pilot in which the only changes required are infrastructure substitutions rather than architectural redesign (Section 4.6).

3 Risks and Challenges

Building a single anomaly detection framework that spans product defects, worker safety, and machinery health introduces risks beyond those of a narrow, single purpose classifier. This section sets out the risks we have identified across the technical, data, and deployment dimensions of the project, together with the mitigation strategy adopted for each.

3.1 Domain Gap Between Public Datasets and Real RMG Conditions

The datasets used to build and evaluate the prototype (public fabric defect datasets, a construction site PPE dataset, and the MVTec-AD industrial anomaly benchmark) were not captured on an actual Bangladeshi garment factory floor. Differences in lighting, camera angle, fabric types, and worker uniform styles mean that accuracy measured on these public datasets will likely be optimistic relative to real deployment conditions.

Mitigation: The competition prototype is positioned as a proof of methodology, with a clearly defined next step of fine tuning on a small set of real factory images obtained through a pilot partnership. Hence, the few-shot design of the underlying model makes it cheap to do (a handful of reference images per new category, not a full retraining run).

3.2 Absence of a Public RMG-Specific Machinery Fault Dataset

No public dataset of sewing machine or garment manufacturing equipment faults exists at the time of writing. For the machinery health category, the prototype substitutes the MVTec-AD industrial benchmark (general mechanical/electronic components) as a proxy to demonstrate that the same architecture generalizes to a third, unrelated object category.

Mitigation: The machinery category is a proof of concept using proxy data, not a claim of RMG machinery specific validation. This is a deliberate, disclosed scope decision rather than an oversight. And it does not affect the validity of the fabric and safety categories, which are backed by domain relevant data.

3.3 Accuracy Ceiling of Zero-Shot/Few-Shot Detection

Zero-shot and few-shot anomaly detection methods generally trade some raw accuracy for the ability to generalize across categories without large labeled datasets per category. Published results for AnomalyCLIP on standard benchmarks (MVTec-AD, VisA) are strong but below what a fully supervised, large dataset classifier trained on one narrow category alone could typically achieve.

Mitigation: This trade-off is treated as the core value proposition. The conclusion drawn in the whitepaper and pitch is ‘Our framework achieves competitive accuracy across three unrelated anomaly categories without needing a separately trained and labeled model for each one’. The generalization property is the innovation being demonstrated.

3.4 Compute Constraints

Fine-tuning a full large vision-language model end to end (the approach used by systems such as AnomalyGPT) requires substantial GPU resources that are not available to a self funded student team. Attempting this under a tight competition deadline risks losing significant time to infrastructure debugging rather than the actual anomaly detection task.

Mitigation: The architecture (Section 4) is deliberately scoped around a lighter CLIP-based backbone (AnomalyCLIP) that does not require full model fine-tuning. Also paired with it is a hosted API (Gemini Flash, free tier) for the natural-language explanation layer rather than self hosting a large multimodal model during the prototype phase. This keeps the entire pipeline runnable on free-tier infrastructure within the competition timeline.

3.5 Cost of False Positives and False Negatives in Real-World Settings

A false negative – an actual defect, hazard, or fault the system misses – carries an obvious cost (a defective shipment, an unaddressed safety risk, an undetected machine fault). A false positive – flagging a compliant scene as anomalous – has a less obvious but real cost. Repeated false alarms erode supervisor trust in the system and risk it being ignored or switched off, which is a common failure mode for real-world anomaly detection deployments generally, not specific to this project.

Mitigation: The system is designed as a supervisor-facing decision support tool with a human in the loop confirmation step, not an automated system that takes action on its own (e.g., it does not automatically reject a fabric roll or lock a machine down). A configurable confidence threshold allows the sensitivity/false-alarm trade-off to be tuned per deployment.

3.6 Privacy and Surveillance Concerns (Safety Category)

Continuous visual monitoring of workers for PPE compliance raises legitimate privacy and workplace surveillance concerns given the sensitivity of worker-monitoring systems generally.

Mitigation: The design processes video/images on-premise where deployed, rather than uploading raw worker imagery to a third-party cloud service by default. Only aggregate compliance statistics and specific flagged frames (not continuous footage) are retained.

3.7 Reliability of the Natural-Language Explanation Layer

The explanation layer uses a vision-language model to describe a detected anomaly in plain language. Like any generative model, it carries some risk of producing a plausible sounding but inaccurate description (a known limitation of LLM/VLM outputs generally), particularly for ambiguous or borderline anomaly regions.

Mitigation: The explanation layer is treated strictly as a secondary, human readable annotation layered on top of the anomaly score and heatmap, not as the basis for the anomaly decision itself. The under-

lying AnomalyCLIP score and localization remain the source of truth, and the explanation is presented alongside the visual heatmap so a human reviewer can independently verify it rather than relying on the text description alone.

3.8 Third-Party API Dependency During Demonstration

The prototype’s explanation layer calls a third-party hosted API (Gemini Flash) rather than a fully self hosted model, which introduces a dependency on external service availability and rate limits, particularly relevant during a live competition demo.

Mitigation: The core anomaly detection function (AnomalyCLIP scoring and localization) runs fully locally and does not depend on the explanation API. So, the system remains demonstrably functional even if the explanation call fails. A cached set of pre-generated explanations for the demo dataset is also prepared as a fallback for the live presentation.

3.9 Risk Summary

Risk	Impact if Unaddressed	Mitigation
Domain gap (public vs. real factory data)	Inflated accuracy claims; poor real-world transfer	Disclosed limitation; few-shot fine tuning path defined
No RMG specific machinery dataset	Machinery category unvalidated for real equipment	MVTec-AD used as disclosed proxy/proof of concept
Zero-shot/few-shot accuracy ceiling	Lower raw accuracy vs. specialized supervised models	Framed and reported as a generalization trade-off
Compute constraints	Risk of losing build time to infrastructure debugging	Lightweight CLIP-based backbone; hosted API for explanation layer
False positive/negative cost	Missed defects/hazards, or alarm fatigue and disuse	Human in the loop design; tunable confidence threshold
Worker privacy/surveillance	Worker trust erosion; ethical/legal exposure	On-premise processing; aggregate-only retention by default
Explanation layer hallucination	Misleading text description of an anomaly	Explanation is secondary annotation only; score/heatmap is source of truth
Third-party API dependency	Demo disruption if API unavailable/rate-limited	Core detection runs locally; cached fallback explanations prepared

Table 2: Risk summary: identified risks, unmitigated impact, and adopted mitigation.

4 Architecture and Infrastructure

This section details the technical design of the RMG Universal Anomaly Detection Framework: the model architecture, the end-to-end inference pipeline, the system/infrastructure design, and the resource and scalability plan from prototype through to a production deployment.

4.1 Design Rationale

Rather than training a separate supervised classifier for each anomaly type (fabric defects, safety violations, machinery faults), the architecture is built around a single, shared vision-language backbone capable of zero-shot/few-shot anomaly detection across arbitrary object categories. This directly satisfies the project’s core requirement – a genuinely universal framework, not three models presented under one interface. And it is grounded in a published, peer-reviewed method rather than a novel and unproven architecture, which keeps technical risk manageable within the competition timeline.

4.2 Model Architecture

The detection backbone is AnomalyCLIP (Zhou et al., ICLR 2024), an object-agnostic prompt learning adaptation of CLIP designed specifically for zero-shot anomaly detection and segmentation across diverse, unseen object categories. Its official public implementation is used as the starting point, adapted with RMG-relevant category prompts and reference images rather than retrained from scratch.

- **Visual encoder:** CLIP ViT-L/14@336px, pretrained and frozen.
- **Learned components:** Lightweight object-agnostic text prompts and a small set of linear projection layers, trained to separate generic ‘normal’ and ‘anomalous’ visual-semantic regions rather than category specific defect types.
- **Output:** An image-level anomaly score (probability the image contains an anomaly) and a pixel-level anomaly heatmap localizing the anomalous region.

Operating mode is few-shot rather than pure zero-shot. For each RMG category (fabric, safety, machinery-proxy), a small bank of 3–5 reference ‘normal’ images is provided at inference time. This will measurably improve detection quality over pure zero-shot prompting while remaining far cheaper than collecting a full labeled training set per category.

4.3 End-to-End Pipeline

Figure 1 shows the full inference flow, from an input image of any of the three RMG categories through to a final verdict and explanation.

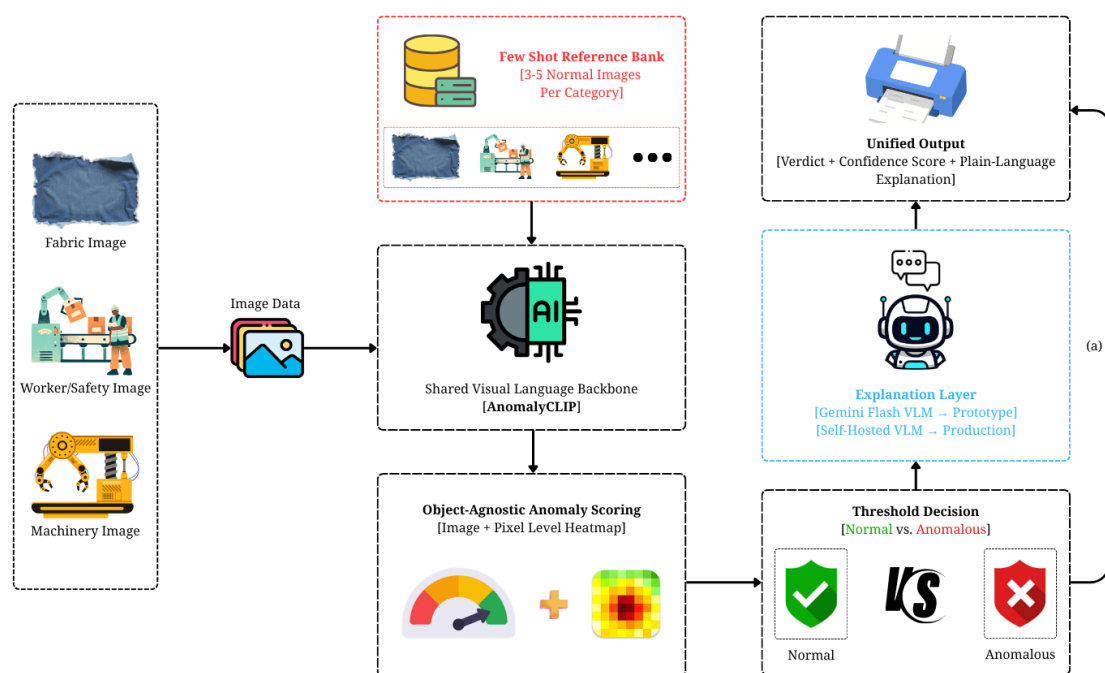


Figure 1: End-to-end inference pipeline — A shared backbone processes any RMG-relevant image category, with the explanation layer invoked only when an anomaly is detected (a).

An uploaded image is passed through the shared AnomalyCLIP backbone alongside the relevant category’s few-shot reference bank. The model produces an anomaly score and heatmap, which are compared against a configurable threshold. If the image is classified as normal, the result is logged directly. If classified as anomalous, the flagged region and score are passed to the explanation layer, which generates a plain language description of the detected issue for the end user.

4.4 Explanation Layer

The explanation layer is intentionally decoupled from the detection layer so the two can evolve independently. For the prototype, it is implemented as a call to Gemini Flash (Google AI Studio’s free tier), chosen for zero-setup integration within the competition timeline. The architecture explicitly supports substituting a self hosted open-weight vision-language model (e.g., Qwen3-VL or LLaVA) for production deployment. This will remove the dependency on an external API and keep factory imagery fully on-premise, directly addressing the privacy consideration raised in Section 3.6.

4.5 System and Infrastructure Architecture

Figure 2 shows the system’s infrastructure layers: a client-facing upload interface, an application layer that orchestrates inference calls, a model-serving layer running the detection and explanation models, and a data/audit layer that stores anomaly logs and the few-shot reference image bank.

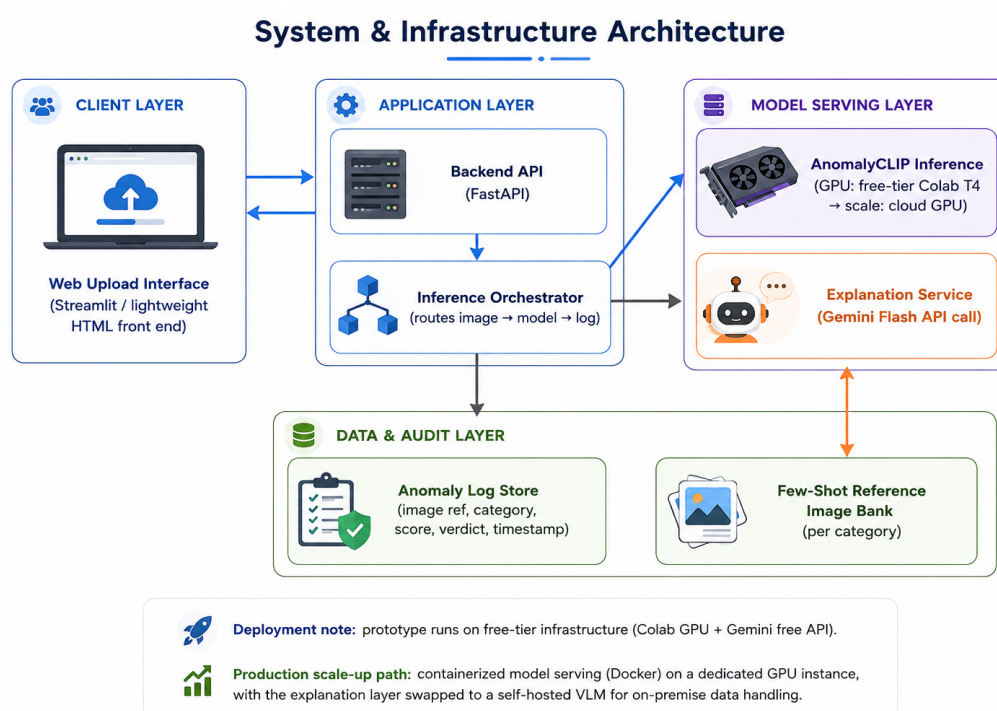


Figure 2: System and infrastructure architecture, showing the prototype’s free-tier deployment and the defined production scale-up path.

- **Client layer:** A lightweight web interface (Streamlit/HTML-CSS for the prototype) for uploading an image and viewing the verdict, heatmap, and explanation.
- **Application layer:** A FastAPI backend exposing a single inference endpoint, with an orchestrator component that routes the image to the detection model, applies the threshold decision, and (if needed) calls the explanation service.
- **Model serving layer:** AnomalyCLIP inference running on a GPU, and a separate explanation service module wrapping the Gemini Flash API call (swappable for a self hosted VLM in production).
- **Data and audit layer:** A lightweight log store recording each inference’s category, score, verdict, and timestamp (supporting the auditability goal from the original problem statement), and the few-shot reference image bank used at inference time.

4.6 Infrastructure Estimation and Scalability

The prototype is designed to run entirely on free-tier infrastructure, consistent with the team’s current budget constraints:

- **Compute:** A single free-tier Colab T4-GPU is sufficient for AnomalyCLIP inference, since the backbone is used in inference/few-shot mode rather than full fine-tuning.
- **Explanation API:** Gemini Flash’s free tier (Google AI Studio) provides sufficient request volume for prototype demonstration and testing at no cost.
- **Storage:** The few-shot reference bank and anomaly log store are small (a few dozen images and lightweight structured records). These are well within free-tier database/storage limits (e.g., SQLite or a free-tier hosted database).

For a production pilot (a single factory partner), the natural scale-up path is: containerizing the model-serving layer (Docker) for deployment on a dedicated GPU instance sized to the factory’s camera count and required inference frequency. Also replacing the hosted explanation API with a self hosted open-weight VLM to remove per request cost and keep imagery on-premise. And finally moving the log store to a managed database sized to the factory’s inspection volume. None of these changes require redesigning the core architecture. They are infrastructure substitutions within the same pipeline shown in Figures 1 and 2, which is a deliberate design choice to keep the prototype to production path low-risk.

4.7 Technology Stack Summary

Component	Technology
Detection backbone	AnomalyCLIP (CLIP ViT-L/14@336px), official public implementation
Explanation layer (prototype)	Gemini Flash API (Google AI Studio, free tier)
Explanation layer (production path)	Self hosted open-weight VLM (e.g., Qwen3-VL / LLaVA)
Backend framework	Python, FastAPI
Frontend/demo interface	Streamlit/HTML-CSS
Training/inference compute	Google Colab free-tier T4-GPU for prototype; dedicated GPU instance for production
Data storage	Lightweight database (e.g., SQLite) for anomaly logs and reference image bank
Version control	GitHub (public repository)

Table 3: Technology stack for the prototype and its defined production path.

5 Revenue and Distribution

5.1 Value and Revenue Model

The framework’s value is generated primarily through cost avoidance and risk reduction for factories: fewer defective shipments reaching buyers, faster safety-issue detection, and a documented audit trail that supports buyer compliance reviews. Direct monetization would follow a per-camera-zone or per-inspection-station subscription model, sold to factory QC and safety departments – comparable in structure to existing industrial-SaaS pricing, and priced well below the imported hardware-based inspection systems described in Section 2.2 since the framework runs on commodity cameras and free-to-low-cost inference infrastructure.

Three properties of the architecture make this model workable rather than aspirational. First, the

marginal cost of covering an additional inspection station is a reference image bank and inference capacity, not a new capital purchase, so pricing can scale smoothly with the number of stations a factory chooses to cover. Second, because the same pipeline serves quality, safety, and machinery screening, a single subscription is sold once and amortised across three departmental budgets that would otherwise each be asked to fund a separate tool. Third, the audit log is a saleable output in its own right: factories already spend staff time assembling compliance evidence for buyer audits, and the system produces that evidence as a by-product of normal operation.

5.2 Cost Structure

Prototype-stage cost is effectively zero, by design (Section 4.6): free-tier GPU compute, a free-tier explanation API, and a lightweight local datastore. At pilot stage the recurring cost is dominated by a single dedicated GPU instance sized to the partner factory's camera count, plus modest managed-database cost. Replacing the hosted explanation API with a self-hosted open-weight VLM converts a per-request variable cost into fixed compute the factory already pays for, which both improves unit economics as inspection volume grows and satisfies the on-premise privacy requirement in Section 3.6. Because the detection backbone is used in inference and few-shot mode rather than being retrained, there is no recurring training cost per customer, per product line, or per new anomaly category.

5.3 Path to Market

- **Phase 1 (Competition prototype):** demonstrate the unified pipeline across fabric, safety, and machinery-proxy categories on public/benchmark data.
- **Phase 2 (Pilot, 3–6 months):** partner with a single factory to fine-tune the few-shot reference bank on real factory imagery and validate accuracy under real conditions.
- **Phase 3 (Sector scale-up):** distribute through BGMEA's existing member network and industry quality-improvement initiatives, and position the tool as support for buyer-compliance audits that factories already need to pass.
- **Phase 4 (Adjacent distribution):** license a white-label version to third-party safety/quality audit firms that already serve the RMG sector, extending reach without factories needing to adopt new vendor relationships directly.

The distribution logic behind this sequence is that the framework does not need to create demand – buyer-compliance auditing is an obligation factories already carry. Phases 3 and 4 therefore attach the product to channels and budget lines that exist today (BGMEA member programmes; incumbent audit firms) rather than requiring factories to establish a new procurement relationship for an unfamiliar category of software.

5.4 Feasibility

Because the core architecture generalizes via few-shot reference images rather than full retraining, adding a new factory, product line, or anomaly category is a low-cost, incremental step rather than a new development project – keeping both onboarding cost and the framework's own operating cost low relative to the buyer-compliance and shipment-quality risks it is designed to reduce.

Feasibility within the competition's own constraints follows the same reasoning. The prototype is built on a published, publicly implemented method (Section 4.2) rather than an unproven architecture, runs entirely on free-tier infrastructure available to a self-funded student team (Section 4.6), and has a scale-up path composed only of infrastructure substitutions inside an unchanged pipeline. The principal open dependency is access to real factory imagery for Phase 2, which the few-shot design deliberately keeps small: a handful of reference images per category, rather than a labelled dataset per task.

References

1. Bangladesh Garment Manufacturers and Exporters Association (BGMEA), “Export Performance,” FY 2025–26, data compiled from the Export Promotion Bureau.
2. J. Jeong, Y. Zou, T. Kim, D. Zhang, A. Ravichandran, and O. Dabeer, “WinCLIP: Zero-/Few-Shot Anomaly Classification and Segmentation,” *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 19606–19616, 2023.
3. Q. Zhou, G. Pang, Y. Tian, S. He, and J. Chen, “AnomalyCLIP: Object-agnostic Prompt Learning for Zero-shot Anomaly Detection,” *International Conference on Learning Representations (ICLR)*, 2024.
4. RMG Sustainability Council (RSC), “Health & Safety,” “Factories,” and “Remediation Programme,” Bangladesh, accessed August 2026.